

Singular Gram limits at the critical line of Dirichlet convolution

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Abstract

Normalized divisibility Gram matrices at the critical coefficient scale $a(k) = k^{-1/2}L(k)$ can have a nonzero entrywise limit and a zero strong-resolvent limit on $\ell^2(\mathbb{N})$. We establish the complete summability dichotomy for positive measurable slowly varying L , without assuming multiplicativity or monotonicity. If $F(N) = \sum_{k \leq N} L(k)^2/k$ diverges, the entries tend to $\gcd(m, n)/\sqrt{mn}$, while the normalized Gram operators converge to zero in strong resolvent sense and their norms tend to infinity. The limiting quadratic form on finitely supported sequences is purely singular: every nonnegative closable minorant vanishes. An explicit average of prime shifts proves this statement and supplies low-energy recovery vectors for arbitrary bounded positive approximants with the same entrywise limit. If F is bounded, the strong-resolvent limit is instead the nonzero operator $C^*C/F(\infty)$, where C is rowwise Dirichlet convolution on its maximal domain. Its proof uses finite-row weak lower bounds and does not assume a finitely supported graph core. For $L(x) = (\log(ex))^{-\beta}$, the two branches meet at $\beta = 1/2$. The conclusions concern a fixed Hilbert space and normalization, and distinguish matrix-entry limits from closed-operator limits.

Keywords: divisibility matrix; GCD kernel; quadratic form; closability; strong resolvent convergence; slow variation.

1 Introduction and the critical dichotomy

A positive limit of finite matrix entries need not define a closed quadratic form on the ambient Hilbert space. Critical divisibility Grams give an arithmetic instance of this distinction: their first diagonal entry stays equal to one, but their resolvents can converge strongly to the resolvent of zero. The issue is therefore not only whether the limiting matrix is bounded, but whether any nonzero closable part survives.

Throughout, $\mathcal{H} = \ell^2(\mathbb{N})$ with its usual coordinate basis (e_m) . Inner products are conjugate-linear in the first argument. Let $L : [1, \infty) \rightarrow (0, \infty)$ be measurable, bounded above and away from zero on compact intervals, and slowly varying:

$$\frac{L(tx)}{L(x)} \rightarrow 1 \quad (x \rightarrow \infty), \quad t > 0. \quad (1.1)$$

Define

$$a(k) = k^{-1/2}L(k), \quad F(x) = \sum_{k \leq [x]} a(k)^2, \quad (1.2)$$

with an empty sum equal to zero. For an integer $N \geq 1$, let

$$T_N(r, m) = \begin{cases} a(r/m), & m \mid r \leq N, \\ 0, & \text{otherwise,} \end{cases} \quad A_N = F(N)^{-1}T_N^*T_N. \quad (1.3)$$

These are finite matrices extended by zero on $\mathcal{H}$, so A_N is bounded, nonnegative and self-adjoint. No uniform bound in N is assumed.

For clarity, a nonnegative form q on $c_{00}(\mathbb{N})$ is *closable* if $u_j \rightarrow 0$ in $\mathcal{H}$ and $q[u_j - u_k] \rightarrow 0$ imply $q[u_j] \rightarrow 0$. A *closable minorant* is a nonnegative closable form b on $c_{00}(\mathbb{N})$ such that $b[u] \leq q[u]$ there. We call q *purely singular* when every such minorant is zero. This refers to form relaxation, not to a singular spectral measure of an already defined self-adjoint operator.

Theorem 1.1 (Critical summability dichotomy). *Under (1.1)–(1.3), exactly one of the following alternatives holds.*

1. If $F(N) \rightarrow \infty$, then for every fixed $m, n \in \mathbb{N}$,

$$\langle e_m, A_N e_n \rangle \rightarrow K(m, n) := \frac{\gcd(m, n)}{\sqrt{mn}}. \quad (1.4)$$

The nonnegative form $q_K[u] = \sum_{m, n} \overline{u(m)} K(m, n) u(n)$ on $c_{00}(\mathbb{N})$ is nonzero and purely singular. It has no nonnegative closed extension agreeing with it on $c_{00}(\mathbb{N})$. Nevertheless, for every $\lambda > 0$,

$$(A_N + \lambda I)^{-1} f \rightarrow \lambda^{-1} f \quad \text{for every } f \in \mathcal{H}, \quad \|A_N\| \rightarrow \infty. \quad (1.5)$$

2. If $F(N) \rightarrow F_\infty < \infty$, define the maximal rowwise Dirichlet convolution operator by

$$(Cx)(r) = \sum_{m|r} a(r/m)x(m),$$

$$D(C) = \left\{ x \in \mathcal{H} : \sum_{r=1}^{\infty} \left| \sum_{m|r} a(r/m)x(m) \right|^2 < \infty \right\}. \quad (1.6)$$

Then C is densely defined and closed. The operator $A_\infty = F_\infty^{-1} C^* C$ is nonnegative, self-adjoint and nonzero, and for every $\lambda > 0$ and $f \in \mathcal{H}$,

$$(A_N + \lambda I)^{-1} f \rightarrow (A_\infty + \lambda I)^{-1} f. \quad (1.7)$$

Here and below, strong-resolvent convergence for nonnegative operators is expressed by the resolvent limits at $-\lambda$, $\lambda > 0$. Theorem 1.1 is one endpoint statement with two alternatives. Its arithmetic ingredient is an explicit prime-tail approximation for the limiting form. The square-summable alternative uses the maximal domain in (1.6), rather than an unproved identification with the graph closure of finitely supported sequences.

Position relative to earlier work. The GCD kernel and its finite positivity are classical. Aistleitner, Berkes and Seip [1] give the finite-product Poisson representation and study finite GCD sums and spectral norms; the role of the prime-square summability threshold already appears there. Hilberdink's arbitrary-coefficient Gram identity [3, Proposition 2.1] supplies the finite algebra underlying (1.3). We rederive that algebra, and do not claim either identity as new.

For power coefficients below the critical line, Hilberdink and Pushnitski [4] obtain compact LCM operators and rescaled-Gram convergence in Schatten classes in their stated range. The comparison here concerns Sections 2.2–2.3 of their 2021 preprint, where the Gram result assumes $\sigma < 1/2$. Theorem 1.1 addresses what replaces a closed limiting operator at the critical coefficient exponent, including slowly varying weights that need not be multiplicative. It does not supply new finite GCD norm asymptotics or a new diagonalization of LCM operators.

General regular/singular form decomposition and the largest closable minorant belong to Simon [5, Section 2]. Our prime-tail vectors identify the purely singular case for the present

arithmetic kernel; their resolvent consequence is a direct variational argument within that classical framework. Simon's monotone convergence theorems are not invoked as an entrywise convergence theorem. Yafaev [6] also emphasizes the distinction between a semibounded matrix form and its closed-operator realization. His one-circle Toeplitz closability criterion is not transferred here to an infinite-prime space.

Section 2 proves the prime-product form criterion. Section 3 converts its recovery vectors into resolvent convergence. Sections 4 and 5 prove the two branches of Theorem 1.1. The final sections give a boundary example, finite verification scopes, and the limits of the fixed-space conclusion.

2 Prime-product forms and their closability boundary

Let (r_p) be numbers indexed by the rational primes with $0 \leq r_p < 1$. Write $v_p(m)$ for the exponent of p in m and put

$$K_r(m, n) = \prod_p r_p^{|v_p(m) - v_p(n)|}, \quad q_r(f, g) = \sum_{m, n} \overline{f(m)} K_r(m, n) g(n), \quad (2.1)$$

for $f, g \in c_{00}(\mathbb{N})$, with $0^0 = 1$ and $q_r[f] = q_r(f, f)$. Only finitely many factors of each product differ from one.

Lemma 2.1 (Finite Poisson positivity). *The form q_r is nonnegative.*

Proof. For $0 \leq r < 1$, the probability density on the unit circle

$$P_r(e^{it}) = \frac{1 - r^2}{|1 - re^{it}|^2}$$

has Fourier coefficients $r^{|j-k|}$:

$$\int_0^{2\pi} e^{i(k-j)t} P_r(e^{it}) \frac{dt}{2\pi} = r^{|j-k|}.$$

This follows by multiplying the two absolutely convergent geometric series; for $r = 0$ the density is one. Given $f \in c_{00}(\mathbb{N})$, choose the finite set S of primes occurring in its support and set $P_f(z) = \sum_m f(m) \prod_{p \in S} z_p^{v_p(m)}$. Integration over $\mathbb{T}^S$ with the product of the displayed probability densities gives $q_r[f] = \int_{\mathbb{T}^S} |P_f(z)|^2 d\mu_S(z) \geq 0$. This is the finite-product representation of Aistleitner et al. [1, Section 3, Lemma 1 in the consulted preprint]. No infinite product measure is required. $\square$

The next identities use primes absent from the support of a vector. Their absence is a hypothesis, not merely a convenient choice of indices. Define the Hilbert-space isometry $S_p e_m = e_{pm}$.

Lemma 2.2 (Prime shifts). *Let $f \in c_{00}(\mathbb{N})$, and let S contain every prime dividing an index in $\text{supp } f$. If $p, \ell \notin S$ are distinct primes, then*

$$\begin{aligned} q_r[S_p f] &= q_r[f], \\ q_r(f, S_p f) &= r_p q_r[f], \\ q_r(S_p f, S_\ell f) &= r_p r_\ell q_r[f]. \end{aligned} \quad (2.2)$$

Moreover, the supports of f and of the vectors $S_p f$ for distinct primes outside S are pairwise disjoint.

Proof. For support indices m, n , their p - and ℓ -adic valuations are zero. Formula (2.1) therefore gives $K_r(pm, pn) = K_r(m, n)$, $K_r(m, pn) = r_p K_r(m, n)$, and $K_r(pm, \ell n) = r_p r_\ell K_r(m, n)$. Summing proves (2.2). If $pm = \ell n$ or $pm = n$ for such indices, unique prime factorization contradicts the asserted absence of p from the relevant index. This proves disjointness. $\square$

Proposition 2.3 (Explicit singularity and converse). *The form q_r is closable on $c_{00}(\mathbb{N})$ if and only if $\sum_p r_p^2 < \infty$. If this sum diverges, q_r is purely singular. More precisely, for every $f \in c_{00}(\mathbb{N})$ there are finite vectors h_j such that*

$$\|h_j\| \rightarrow 0, \quad q_r[h_j - f] \rightarrow 0. \quad (2.3)$$

Consequently there are finite vectors $u_j \rightarrow f$ in $\mathcal{H}$ with $q_r[u_j] \rightarrow 0$.

Proof. Suppose first that $\sum_p r_p^2 = \infty$. Fix f and a finite prime set S as in Lemma 2.2. For a finite set E of primes outside S , write

$$s_E = \sum_{p \in E} r_p^2 > 0, \quad h_E = s_E^{-1} \sum_{p \in E} r_p S_p f.$$

Disjoint supports and (2.2) yield the exact formulas

$$\|h_E\|^2 = \frac{\|f\|^2}{s_E}, \quad q_r(f, h_E) = q_r[f], \quad (2.4)$$

$$q_r[h_E] = q_r[f] \left(1 + \frac{\sum_{p \in E} r_p^2 (1 - r_p^2)}{s_E^2} \right). \quad (2.5)$$

Indeed, the diagonal terms of the double sum for $q_r[h_E]$ contribute $q_r[f] s_E / s_E^2$, and the off-diagonal terms contribute $q_r[f] (s_E^2 - \sum_{p \in E} r_p^4) / s_E^2$. Subtracting the mixed terms gives

$$q_r[h_E - f] = q_r[f] \frac{\sum_{p \in E} r_p^2 (1 - r_p^2)}{s_E^2} \leq \frac{q_r[f]}{s_E}. \quad (2.6)$$

Removing the finite set S does not affect divergence, so choose finite E_j with $s_{E_j} \rightarrow \infty$. Equations (2.4) and (2.6) prove (2.3).

Let $b \leq q_r$ be a nonnegative closable minorant. The triangle inequality for the seminorm $\sqrt{b[\cdot]}$ shows that $b[h_{E_j} - h_{E_k}] \rightarrow 0$, because both vectors approximate f in that seminorm. Since $h_{E_j} \rightarrow 0$ in $\mathcal{H}$, closability gives $b[h_{E_j}] \rightarrow 0$. Applying the triangle inequality once more, and using $b[h_{E_j} - f] \leq q_r[h_{E_j} - f] \rightarrow 0$, gives $b[f] = 0$. This holds for every finite f , so the form is purely singular. It is not zero, since $q_r[e_1] = 1$, and hence cannot itself be closable. Setting $u_j = f - h_{E_j}$ proves the recovery assertion.

Conversely, suppose $\sum_p r_p^2 < \infty$. For a fixed m and $a_p = v_p(m)$, nonnegative summation over prime factorizations gives

$$\sum_{n=1}^{\infty} |K_r(m, n)|^2 = \prod_{p|m} \left(\sum_{j=0}^{\infty} r_p^{2|a_p-j|} \right) \prod_{p \nmid m} (1 - r_p^2)^{-1} < \infty. \quad (2.7)$$

The first product is finite and each of its factors is a finite sum plus a convergent geometric tail. For the second product, all but finitely many r_p^2 are at most $1/2$, and $-\log(1 - r_p^2) \leq 2r_p^2$ there. This proves its convergence; zero parameters cause no exception.

Thus each column of K_r belongs to $\mathcal{H}$, and finite linear combinations define an operator Q_0 on $c_{00}(\mathbb{N})$. The symmetry of the kernel and Lemma 2.1 make Q_0 a densely defined nonnegative symmetric operator. The quadratic form of such an operator is closable, a standard semibounded-form fact recalled in Yafaev [6, Section 2.1]. Its form on $c_{00}(\mathbb{N})$ is exactly q_r , proving the converse. $\square$

A nonnegative closed extension agreeing with the divergent-case form on $c_{00}(\mathbb{N})$ would have a nonzero closable restriction to $c_{00}(\mathbb{N})$. Proposition 2.3 rules this out. The proposition supplies an explicit witness within Simon's regular/singular framework [5]; it does not introduce a new general decomposition theorem.

3 A variational consequence for positive approximants

The recovery vectors in Proposition 2.3 carry more information than a single failure of closability. They approximate any specified finite vector with arbitrarily small energy, which determines the resolvent limit of every positive entrywise approximation.

Proposition 3.1. *Assume $\sum_p r_p^2 = \infty$. Let B_N be bounded nonnegative self-adjoint operators on $\mathcal{H}$ such that*

$$\langle e_m, B_N e_n \rangle \longrightarrow K_r(m, n) \quad \text{for every fixed } m, n.$$

Then $(B_N + \lambda I)^{-1} \rightarrow \lambda^{-1} I$ strongly for every $\lambda > 0$, and $\|B_N\| \rightarrow \infty$. No monotonicity or uniform norm bound on the approximants is required.

Proof. For each fixed $v \in c_{00}(\mathbb{N})$, finite summation gives

$$\langle v, B_N v \rangle \longrightarrow q_r[v]. \quad (3.1)$$

Let $g \in \mathcal{H}$. Choose finite $f_j \rightarrow g$ and then, by Proposition 2.3, choose finite v_j with $\|v_j - f_j\| < 1/j$ and $q_r[v_j] < 1/j$. For each j , (3.1) supplies an integer N_j such that

$$|\langle v_j, B_N v_j \rangle - q_r[v_j]| < 1/j \quad \text{for all } N \geq N_j.$$

Choose the N_j strictly increasing and tending to infinity. For $N \geq N_1$, let $j(N) = \max\{j : N_j \leq N\}$. The finitely many smaller N can be assigned any finite vector. Then

$$w_N := v_{j(N)} \longrightarrow g, \quad \langle w_N, B_N w_N \rangle \longrightarrow 0. \quad (3.2)$$

This diagonal choice is what permits a moving support; entrywise convergence alone was used only for each fixed v_j .

Fix $\lambda > 0$ and $f \in \mathcal{H}$ and put $x_N = (B_N + \lambda I)^{-1} f$. The unique minimizer of

$$J_N(x) = \langle x, B_N x \rangle + \lambda \|x - \lambda^{-1} f\|^2$$

is x_N , as follows by expanding the square and using the resolvent equation. Apply (3.2) to $g = \lambda^{-1} f$. Positivity and minimality imply

$$0 \leq \lambda \|x_N - \lambda^{-1} f\|^2 \leq J_N(x_N) \leq J_N(w_N) \longrightarrow 0.$$

This proves the strong-resolvent assertion.

Finally q_r is unbounded on the finite unit vectors: otherwise it would extend to a bounded nonnegative form on $\mathcal{H}$, making its restriction a nonzero closable minorant of itself. For each $M > 0$ choose a fixed finite unit vector u with $q_r[u] > 2M$. Equation (3.1) gives $\|B_N\| \geq \langle u, B_N u \rangle > M$ for all sufficiently large N . Thus the norms tend to infinity, not merely along a subsequence. $\square$

The proposition is a direct relaxation and minimization argument. It does not apply Simon's monotone form-convergence theorems [5, Section 3] to a nonmonotone family. It also makes no assertion about arbitrary signed approximants: nonnegativity is used in the decisive minimizer estimate.

4 Critical regular variation and the divergent branch

We now identify the entrywise limit in Theorem 1.1. The only regular-variation input is the classical uniform convergence theorem for positive measurable slowly varying functions [2]: for each compact $J \subset (0, \infty)$,

$$\sup_{t \in J} \left| \frac{L(tx)}{L(x)} - 1 \right| \longrightarrow 0 \quad (x \rightarrow \infty). \quad (4.1)$$

We state the version used explicitly, without requiring monotonicity of L or convergence of L itself.

Lemma 4.1 (Critical partial sums). *If $F(N) \rightarrow \infty$, then*

$$L(N)^2 = o(F(N)), \quad \frac{F(N/c)}{F(N)} \longrightarrow 1 \quad \text{for each fixed } c \geq 1. \quad (4.2)$$

Proof. Fix $A > 1$. Uniform convergence on $[1/A, 1]$ gives

$$\frac{1}{L(N)^2} \sum_{\lceil N/A \rceil \leq k \leq N} \frac{L(k)^2}{k} \longrightarrow \log A, \quad (4.3)$$

since the harmonic sum on the same interval tends to $\log A$. Hence $\liminf_{N \rightarrow \infty} F(N)/L(N)^2 \geq \log A$. Only after this limit do we let A grow: for every fixed bound M , choose A with $\log A > M$. It follows that $F(N)/L(N)^2 \rightarrow \infty$.

For fixed $c > 1$, the omitted indices in $F(N) - F(N/c)$ satisfy $N/c < k \leq N$. By (4.1), their weights $L(k)^2$ are at most a fixed multiple of $L(N)^2$ for all sufficiently large N , and $\sum_{N/c < k \leq N} 1/k$ is bounded in N . Thus $0 \leq F(N) - F(N/c) = O_c(L(N)^2) = o(F(N))$. Floors change only the interval endpoints and are already included in these sums. The case $c = 1$ is immediate. $\square$

Lemma 4.2 (Finite Gram identity). *For fixed m, n and $\ell = \text{lcm}(m, n)$, if $N \geq \ell$ then*

$$\langle e_m, A_N e_n \rangle = \frac{\sqrt{mn}}{\ell F(N)} \sum_{k \leq N/\ell} \frac{L(k\ell/m)L(k\ell/n)}{k}. \quad (4.4)$$

If $N < \ell$, the entry is zero.

Proof. A row contributes to $\langle T_N e_m, T_N e_n \rangle$ precisely when it is a common multiple of m and n . Write that row as $k\ell$. Its contribution is

$$a(k\ell/m)a(k\ell/n) = \frac{\sqrt{mn}}{k\ell} L(k\ell/m)L(k\ell/n).$$

Summing over $k\ell \leq N$ and dividing by $F(N)$ proves the claim. This is the specialization of the arbitrary-coefficient identity in Hilberdink [3, Proposition 2.1]. $\square$

Proof of Theorem 1.1, divergent branch. Fix m, n , and set $\ell = \text{lcm}(m, n)$. The two factors ℓ/m and ℓ/n are fixed positive integers, so slow variation implies

$$\frac{L(k\ell/m)L(k\ell/n)}{L(k)^2} \longrightarrow 1.$$

For any $\varepsilon > 0$, this ratio lies between $1 - \varepsilon$ and $1 + \varepsilon$ for all k beyond a fixed index. Split the sum in (4.4) at that index. The initial part is finite, whereas $F(N/\ell) \rightarrow \infty$. Consequently

$$\sum_{k \leq N/\ell} \frac{L(k\ell/m)L(k\ell/n)}{k} \sim F(N/\ell) \sim F(N),$$

the second equivalence following from Lemma 4.1. Formula (4.4) therefore tends to $\sqrt{mn}/\ell = \gcd(m, n)/\sqrt{mn}$.

Unique prime factorization identifies this kernel as

$$K(m, n) = \prod_p \left(p^{-1/2} \right)^{|v_p(m) - v_p(n)|}.$$

The classical divergence of $\sum_p 1/p$ gives the hypothesis of Propositions 2.3 and 3.1. They establish pure singularity, the nonexistence of a nonnegative closed extension, (1.5), and norm divergence. $\square$

The diagonal identity

$$\langle e_1, A_N e_1 \rangle = F(N)^{-1} \sum_{r \leq N} a(r)^2 = 1 \quad (4.5)$$

holds for every N , in both branches. In the divergent branch it precludes strong operator convergence $A_N \rightarrow 0$, since such convergence would force the displayed scalar to tend to zero. Strong-resolvent convergence in (1.5) is a different assertion and is compatible with (4.5).

5 The square-summable alternative on the maximal domain

When $F_\infty < \infty$, the sequence a is in $\ell^2(\mathbb{N})$. The formal row formula in (1.6) still makes sense for every $x \in \mathcal{H}$, because each integer has only finitely many divisors. The condition defining $D(C)$ asks whether these row values form an ℓ^2 sequence; it is not a condition on limits of coordinate truncations in a graph norm.

Lemma 5.1 (Closed maximal convolution). *If $a \in \ell^2(\mathbb{N})$, the operator C in (1.6) is closed and densely defined. Its nonnegative closed form*

$$q_\infty[x] = F_\infty^{-1} \|Cx\|^2, \quad x \in D(C), \quad (5.1)$$

is associated with $A_\infty = F_\infty^{-1} C^ C$.*

Proof. Suppose $x_j \rightarrow x$ and $Cx_j \rightarrow y$ in $\mathcal{H}$. For each fixed row r , the map $x \mapsto \sum_{m|r} a(r/m)x(m)$ is a continuous linear functional, being a finite sum of coordinates. Passing to the limit gives $(Cx)(r) = y(r)$ for every r . Thus the formal sequence Cx equals $y \in \mathcal{H}$, which proves $x \in D(C)$ and closedness.

For each m , the column Ce_m has values $a(k)$ at the indices mk and zero elsewhere, so $\|Ce_m\|^2 = \|a\|^2 = F_\infty$. Finite linear combinations show $c_{00}(\mathbb{N}) \subset D(C)$, hence density. Since C is closed, its graph norm is complete; this proves that (5.1) is a densely defined closed form. The standard representation for the norm-square form of a closed operator identifies its associated nonnegative self-adjoint operator as $F_\infty^{-1} C^* C$. $\square$

For every $x \in \mathcal{H}$, set $q_N[x] = \langle x, A_N x \rangle$. The finite matrices give the exact identity

$$q_N[x] = F(N)^{-1} \sum_{r \leq N} \left| \sum_{m|r} a(r/m)x(m) \right|^2. \quad (5.2)$$

Indeed, a coordinate with index greater than N cannot divide a row at most N . For a fixed $x \in D(C)$,

$$q_N[x] \longrightarrow q_\infty[x]. \quad (5.3)$$

No monotonicity of the *normalized* forms q_N is needed.

Lemma 5.2 (Finite-row weak lower bound). *If $x_j \rightharpoonup x$ in $\mathcal{H}$ and $N_j \rightarrow \infty$, then*

$$q_\infty[x] \leq \liminf_{j \rightarrow \infty} q_{N_j}[x_j], \quad (5.4)$$

where $q_\infty[x] = +\infty$ for $x \notin D(C)$.

Proof. Fix an integer R . For all sufficiently large j , $N_j \geq R$. Each of the first R row functionals is weakly continuous, and $F(N_j) \rightarrow F_\infty > 0$. Dropping the nonnegative remaining rows in (5.2) gives

$$F_\infty^{-1} \sum_{r \leq R} \left| \sum_{m|r} a(r/m)x(m) \right|^2 \leq \liminf_j q_{N_j}[x_j].$$

Letting $R \rightarrow \infty$ proves (5.4), including its infinite value outside the maximal domain. $\square$

Proof of Theorem 1.1, summable branch. Fix $f \in \mathcal{H}$ and $\lambda > 0$. Write

$$x_N = (A_N + \lambda I)^{-1}f, \quad x_* = (A_\infty + \lambda I)^{-1}f.$$

The resolvent bound gives $\|x_N\| \leq \lambda^{-1}\|f\|$. Consider the functionals

$$E_N(x) = q_N[x] + \lambda\|x\|^2 - 2\operatorname{Re}\langle f, x \rangle, \quad E_\infty(x) = q_\infty[x] + \lambda\|x\|^2 - 2\operatorname{Re}\langle f, x \rangle,$$

with $E_\infty = +\infty$ off $D(C)$. Their unique minimizers are x_N and x_* , respectively. The latter assertion follows from the closed-form representation in Lemma 5.1; in particular $x_* \in D(C)$. Equation (5.3) gives $E_N(x_*) \rightarrow E_\infty(x_*)$.

Take any weakly convergent subsequence $x_{N_j} \rightharpoonup x$. The weak lower bound of Lemma 5.2, weak lower semicontinuity of the norm, and weak continuity of the last linear term yield

$$\begin{aligned} E_\infty(x) &\leq \liminf_j E_{N_j}(x_{N_j}) \\ &\leq \limsup_j E_{N_j}(x_{N_j}) \leq E_\infty(x_*). \end{aligned} \quad (5.5)$$

The last inequality uses minimality against the fixed competitor x_* . It follows that $x \in D(C)$ and $x = x_*$ by uniqueness. Every subsequence has a weakly convergent further subsequence, since the sequence is bounded in the separable Hilbert space $\mathcal{H}$. Thus $x_N \rightharpoonup x_*$ for the full sequence. Applying the lower bound to this sequence and retaining the same upper comparison proves

$$E_N(x_N) \longrightarrow E_\infty(x_*).$$

Finally the resolvent equation and expansion of E_N give

$$\begin{aligned} E_N(x_*) - E_N(x_N) &= q_N[x_* - x_N] + \lambda\|x_* - x_N\|^2 \\ &\geq \lambda\|x_* - x_N\|^2. \end{aligned} \quad (5.6)$$

Both functional values on the left tend to $E_\infty(x_*)$, so $x_N \rightarrow x_*$ in norm, proving (1.7).

The limiting form is nonzero, because $q_\infty[e_1] = F_\infty^{-1}\|a\|^2 = 1$. The uniqueness of its associated operator implies $A_\infty \neq 0$. This argument uses $e_1 \in D(C)$; it does not assume $e_1 \in D(C^*C)$. $\square$

The proof recovers at the actual limiting minimizer x_* . It therefore does not require $c_{00}(\mathbb{N})$ to be a graph core for C , does not identify the maximal operator with the closure of an unstated initial operator, and does not assert essential self-adjointness.

6 Boundary examples, finite checks and scope

The logarithmic family already realizes both alternatives of the theorem. For $\beta \in \mathbb{R}$, let

$$L_\beta(x) = (\log(ex))^{-\beta}, \quad a_\beta(k) = \frac{1}{\sqrt{k}(\log(ek))^\beta}.$$

These functions satisfy all hypotheses on L . The positive function $1/(x(\log(ex))^{2\beta})$ is eventually decreasing. Integral comparison and the substitution $t = \log(ex)$ give

$$\int_1^\infty \frac{dx}{x(\log(ex))^{2\beta}} = \int_1^\infty t^{-2\beta} dt.$$

Therefore $F_\infty < \infty$ exactly when $\beta > 1/2$. At the boundary $\beta = 1/2$ the sum still diverges. For $\beta \neq 0$, $a_\beta(6) \neq a_\beta(2)a_\beta(3)$ because $(1 + \log 2)(1 + \log 3) > 1 + \log 6$. Thus these examples include genuinely nonmultiplicative coefficients.

	$F(N) \rightarrow \infty$	$F(N) \rightarrow F_\infty < \infty$
Entrywise limit	Universal kernel $\gcd(m, n)/\sqrt{mn}$	Gram form of the maximal convolution, divided by F_∞
Closed limiting object	Zero strong-resolvent limit; entrywise form purely singular	Nonzero C^*C/F_∞ on its operator domain
First diagonal entry	$A_N(1, 1) = 1$ for all N	$A_N(1, 1) = 1$ for all N
Logarithmic example	$\beta \leq 1/2$	$\beta > 1/2$

Table 1: The two critical alternatives on the same space $\ell^2(\mathbb{N})$ and with the same normalization $F(N)$. In the summable column, the Gram form means $\langle Ce_m, Ce_n \rangle / F_\infty$, not an assertion that each e_m belongs to $D(C^*C)$.

Exact finite sentinels. The accompanying finite checks use rational arithmetic after the coordinate change $f(m) = b(m)\sqrt{m}$, with $b(1) = 1$, $b(2) = -1$, $b(3) = 1/2$, $b(6) = 1/3$ and zero otherwise. At $r_p = p^{-1/2}$ this makes

$$q_K[f] = \sum_{m,n} b(m)b(n) \gcd(m, n)$$

rational. The prime tails $\{5, 7, 11\}$ and $\{5, 7, 11, 13, 17, 19\}$ satisfy the hypotheses of Lemma 2.2. The checks verify (2.4)–(2.6) exactly. For the first tail, for example, $s_E = 167/385$ and $q_K[h_E - f] = 148819/27889$. This residual is not small; the finite check tests an identity, not proximity to the asymptotic regime.

For the invalid tail $\{2, 3, 5\}$, the norm and mixed-form identities both fail. The support primes overlap the tail, and the shifted supports can collide at index 6. This control tests the actual missing hypothesis. The verification receipt also records an earlier ineffective negative-control choice; it is not reported as a successful control.

For $L = 1$, write $H_j = \sum_{k \leq j} 1/k$, with $H_0 = 0$. Formula (4.4) becomes

$$\langle e_m, A_N e_n \rangle = \frac{\gcd(m, n)}{\sqrt{mn}} \frac{H_{\lfloor N/\text{lcm}(m, n) \rfloor}}{H_N}.$$

Independent rowwise and pairwise evaluations agree exactly for $N = 12, 31, 100$. At $N = 12$, the energy of the displayed finite vector is 374167/172042. These checks concern finite identities only. They do not evaluate resolvents, establish a uniform convergence theorem, or certify an infinite prime tail. All limiting conclusions above follow from proofs.

Limits of the conclusion. Finite positivity alone neither proves nor disproves closability. For example, when $\sum_p r_p^2 < \infty$, Proposition 2.3 gives a positive closable form, and its divergent-tail argument is unavailable. In Theorem 1.1, the summability branch must likewise be decided rather than inferred from the critical power $k^{-1/2}$ alone.

The divergent-case obstruction concerns nonnegative closed forms agreeing with K on $c_{00}(\mathbb{N})$ inside the specified unweighted ℓ^2 space. It does not exclude changes of Hilbert space, other normalizations, indefinite constructions, or unrelated spectral representations. No target Euler factors, root numbers, zero correspondence, or Hilbert–Pólya realization follow from these arithmetic Gram limits.

7 Conclusion and reproducibility

At the critical Dirichlet coefficient exponent, the normalization $F(N)$ separates two different closed-operator limits. Divergence gives a universal nonzero entrywise GCD form, but the explicit prime-tail vectors annihilate every closable minorant and force positive approximants to converge to zero in strong resolvent sense. Square summability gives a nonzero maximal-convolution limit. The distinction is between types of limiting objects, not between two constants in a common norm-convergence formula.

The complete arguments are contained in this article. The accompanying `exact_checks.py` uses Python rational arithmetic to verify only the finite identities in Section 6. Its command, outputs and failed-control history appear in `BOUNDED_RECEIPTS.md`. No random sampling, GPU computation or numerical resolvent extrapolation is part of the proof.

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